Track 1 Report — Submission 1: Exomiser baseline on the provided VCF

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Submission file: `smarin00_exomiser-vendorvcf.csv`
Code: https://github.com/smarin00/mva-hackathon-2026 (private during the hackathon; public after close)
Date: 6 October 2026

1. Summary

This first submission is a deliberately conservative baseline: phenotype-driven prioritisation of the provided vendor VCF with Exomiser, with no re-alignment or re-calling. Its purpose is to establish a reproducible reference point against which the read-level analyses planned for later submissions can be compared.

Without any gene-level prior, Exomiser ranked **BUB1B** first under both recessive and dominant models, well separated from the next gene. Biallelic BUB1B variants cause Mosaic Variegated Aneuploidy type 1 (MVA1), so the top-ranked gene matches the clinical diagnosis. The primary prediction is the two BUB1B candidate alleles, a stop-gained and a rare predicted-damaging missense variant, submitted together as one compound-heterozygous row. This truncating-plus-missense combination matches the genotype pattern reported in MVA1. One secondary finding (LZTR1) is also reported. Other Exomiser candidates were excluded after documented review.

2. Data

Input	Description
Variant calls	Provided VCF: GATK calls with hard filtering (VariantFiltration), GRCh38 analysis set with hs38d1 decoys, unprefixed contigs; ~5.0 M records, 94.6% PASS
Phenotype	HPO terms supplied in the clinical phenotype document, each verified against the current HPO release with `pyhpo`
Raw reads	4-lane paired-end FASTQ (not used in this submission; see §7)

Non-PASS records were reviewed by filter category. Most fail on low mapping quality (MQ < 40), which clusters in segmental duplications and paralogous regions. They were excluded here but kept in view for targeted review of candidate genes later.

3. Pre-processing

1. **Contig naming:** restricted to primary chromosomes (chr1–22, X, Y, M) and renamed to chr-prefixed names, matching the submission format and the annotation resources. The 2.3% of records on decoy and unplaced contigs were dropped.
2. **Normalisation** (bcftools): PASS only; multiallelic records split; indels left-aligned against GRCh38 so that representations match population and clinical databases and the answer key.

4. Prioritisation

Tool: Exomiser 15.1.0 (native jar, Java 21), data release 2602 (Ensembl 112 MANE transcripts; gnomAD v4.1; dbNSFP 4.5a; ClinVar 2026-02-08; HPO 2026-01-08).

Setting	Value	Rationale
Analysis mode	PASS_ONLY	Quality-filtered calls only
Variant effects kept	Coding and splice-region	Memory-bounded genome analysis; deep intronic variants handled separately (§7)
Max population frequency	2% (all gnomAD populations, UK10K)	Permissive for a recessive model, where carrier frequencies can be relatively high
Pathogenicity sources	REVEL, MVP, AlphaMissense, SpliceAI	Complementary missense and splice predictors
Inheritance modes	All modes considered; compound-het AR weighted highest	MVA is autosomal recessive in all known forms
Prioritisers	hiPHIVE (human, mouse, fish phenotypes + PPI) and OMIM	Phenotype similarity drives gene ranking

A fresh ClinVar release (2026-10-04) was prepared for the VEP annotation track, used to review candidates in later submissions.

5. Submission construction and EPCR

The ranked Exomiser variants were converted to the submission format with a scripted, validated step (`bin/make_submission.py`) rather than by hand:

- **Compound-het pairs on one row.** If the answer is a compound heterozygote, only a row that contains both alleles can score a full match. Pairs were built only from variants Exomiser scored under a recessive mode. Dominant-only hits stay as single rows.
- **Hedging.** Each allele of the leading pair is also listed as a single row, below the pair, to retain partial credit if only one allele is correct.
- **Artefact-prone loci excluded:** HLA, KIR, MUC, olfactory-receptor and immunoglobulin/TCR genes, where short-read calls are dominated by mapping error.
- **ClinVar-benign variants excluded** automatically, whatever their Exomiser rank.
- **No pairing of nearby variants.** Two variants less than 50 bp apart are treated as one event on the same haplotype (complex indels split into separate records, alternative repeat lengths at one site), not as a compound heterozygote.
- **No lone heterozygous recessive hits.** A single heterozygous variant that fits only a recessive model cannot explain the disease, so it is submitted only as a hedge for a gene that has a valid pair.
- **Minimum gene score** (Exomiser combined score ≥ 0.1) for primary rows. Unused rows stay empty rather than being filled with low-evidence candidates.
- **Documented gene-level curation** (`docs/curation.tsv`): each Exomiser candidate below the leading gene was reviewed against its consequence, population frequency, ClinVar status and disease association, and either excluded or reclassified as a secondary finding, with the reason recorded. The script applies these decisions, so the submission stays reproducible.
- **Secondary findings ranked last**, with EPCRs below every primary row, so they can never count as false positives against the primary prediction.
- **EPCR:** strictly distinct and descending. Values follow the rank order of the Exomiser scores and **are not calibrated probabilities.**
- **Validation:** before writing, the script checks the same constraints as the Space's scorer (columns, 1–10 rows, EPCR in (0, 1], valid `finding_type`, chr-prefixed contigs), and confirms every submitted variant exists exactly as written in the normalised VCF.

6. Results

6.1 Exomiser gene ranking

Exomiser rank	Gene	Mode	Combined score	Phenotype score
1	BUB1B	AD	0.587	0.564
2	BUB1B	**AR**	0.554	**0.792**
3	FANCD2	AR	0.214	—

BUB1B leads by a wide margin. The AD and AR rows describe the same two alleles: the AD score is driven by the single strongest variant, while the much higher phenotype score under AR reflects the match between the proband's HPO profile and MVA1, a recessive disorder.

6.2 Submitted predictions

Rank	Row	Gene	EPCR	Consequence	Max pop. freq	Pathogenicity	ClinVar	Finding
1	pair	BUB1B	0.95	stop gained + missense	0.0100% + 0.0001%	— + AlphaMissense 0.92	P/LP + not in ClinVar	primary
2	single	BUB1B	0.70	stop gained	0.0100%	—	P/LP	primary (hedge)
3	single	BUB1B	0.45	missense	0.0001%	AlphaMissense 0.92	not in ClinVar	primary (hedge)
4	single	LZTR1	0.01	stop gained	0.0030%	—	P/LP	**secondary**

All variants are heterozygous. The two BUB1B variants lie about 11 kb apart.

BUB1B: primary prediction. BUB1B encodes BUBR1, a core component of the spindle assembly checkpoint, which delays anaphase until every chromosome is attached to the spindle. Biallelic BUB1B variants cause MVA1 (Hanks et al., 2004); a weakened checkpoint lets chromosomes missegregate, producing the mosaic, variegated aneuploidy that defines the disorder.

The proband carries two heterozygous BUB1B variants that fit the reported MVA1 genotype:

- a **stop-gained** allele, very rare (0.01%) and already classified pathogenic/likely pathogenic in ClinVar: a loss-of-function allele;
- a **missense** allele, extremely rare (0.0001%), not in ClinVar, and predicted damaging (AlphaMissense 0.92): a candidate **hypomorphic** allele.

Reported MVA1 genotypes commonly combine one truncating allele with one allele that reduces, rather than abolishes, BUBR1 function, consistent with complete loss being incompatible with development. This truncating-plus-missense configuration is therefore the expected one. Two points remain open: whether the two variants lie on different chromosome copies (trans), and the functional effect of the missense change.

LZTR1: secondary finding. A heterozygous stop-gained variant, classified pathogenic/likely pathogenic in ClinVar. Heterozygous LZTR1 loss-of-function is associated with predisposition to schwannomatosis, with incomplete penetrance. This finding is unrelated to the MVA phenotype and is reported as secondary, ranked below all primary rows.

6.3 Candidates excluded on review

Gene	Exomiser rank	Reason for exclusion
FANCD2	3	Both variants (splice donor, splice region) classified benign/likely benign in ClinVar; FANCD2 pseudogene makes short-read calls unreliable
HK1	lower	Intronic variant at 1.3% population frequency, too common for a rare disorder; heterozygous carrier state only
ADAMTS1	lower	Two frameshifts with identical population frequency, suggesting a single cis haplotype rather than compound heterozygosity; no established Mendelian disease
MTCH2	lower	No established Mendelian disease; low phenotype score
HLA-DRB1	lower	Hypervariable locus; calls dominated by mapping artefacts
SERPINA1, PCDHA4, LILRB1, PCDHB8	lower	"Pairs" 2–5 bp apart: one complex indel split into two records, same haplotype
SLU7	lower	Two insertion lengths at one poly-A site: repeat-length noise
KCNJ18	lower	Segmentally duplicated gene (KCNJ12/KCNJ18); unreliable short-read mapping

Removing these candidates does not change the automated score. It keeps the submitted list limited to findings that survive expert review.

7. Limitations and next steps

Limitation in this submission	Planned remedy
Vendor calls only (GATK hard filtering)	Re-alignment from FASTQ (bwa-mem, GRCh38 analysis set) and independent re-calling with DeepVariant; caller concordance reported
Phase unknown (singleton, no parents)	Read-backed phasing (WhatsHap) of the BUB1B pair. The variants are ~11 kb apart, beyond a single read pair, so phase can only be resolved if intervening heterozygous SNPs chain them into one block; otherwise it remains formally unresolved
Missense allele effect is predicted only	Structural context (AlphaFold model of BUBR1, domain location), conservation, and literature on reported MVA1 missense alleles
Coding/splice-region only	Rare intronic variants in a ~100-gene SAC/kinetochore/centromere panel scored with SpliceAI and Pangolin; genome-wide sweep of rare intronic variants; "second hit" search in any gene with one strong allele
Germline caller assumptions	Mosaic-aware calling (Mutect2) across the candidate panel
SNVs/indels only	Structural variants (Delly) and repeat expansions (ExpansionHunter)
No direct link to the cellular phenotype	Read-depth and B-allele-frequency profiling (CNVpytor) to detect and quantify mosaic aneuploidy from the patient's own reads

Predicted splicing effects remain predictions. Confirming them would need patient RNA or a minigene assay.

8. Reproducibility

- **Workflow:** Nextflow 26.04.6; this submission is `nextflow run main.nf -params-file params.yaml --mode vcf`.
- **Versions:** bcftools/samtools; Exomiser CLI 15.1.0 + data 2602; VEP 116.2 (pinned local container) with cache 116 and plugins from VEP_plugins release/116; full list in `docs/reference_data.md` and `results/pipeline_info/versions.txt`.
- **Python environment:** uv-managed, locked in `requirements.lock`.
- **Decisions log:** `docs/decisions.md` records analysis choices and their rationale.

9. Data handling

All analysis ran on a dedicated, key-only cloud server. No genotype-level data is held in the code repository (history verified). Deletion of all data and derived genotype-level files is planned and tracked in `docs/DELETION.md`, following the organisers' guidance in Space Discussion #2.

LLM use: Anthropic Claude via claude.ai (`Pro` plan, consumer terms), model-training setting disabled ("Help improve Claude" off), no output ratings given. Used for pipeline design, code, troubleshooting and report drafting. No genome-scale data was shared with it.

References

- Hanks S, et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. *Nat Genet* 2004;36:1159–1161.
- Smedley D, et al. Next-generation diagnostics and disease-gene discovery with the Exomiser. *Nat Protoc* 2015;10:2004–2015.
- Stenton SL, et al. Critical assessment of variant prioritization methods for rare disease diagnosis within the Rare Genomes Project. *Hum Genomics* 2024 (CAGI6 RGP; basis of the Track 1 scoring).

Appendix: Methods description form (Track 1, model 1)

Methods description — MVA Hackathon 2026, Track 1 (Variant Prediction). Answers as submitted in the organisers' methods description template.

Team name	smarin00
Model number (fill out this form once per model/approach submitted; up to 6)	1 (Exomiser baseline on the provided VCF)
Please describe your model/approach in detail.	Phenotype-driven prioritisation of the provided GATK VCF, implemented as a Nextflow pipeline (mode 'vcf'). 1) Pre-processing (bcftools): restricted to primary chromosomes and renamed to chr-prefixed contigs; PASS only; multiallelic records split; indels left-aligned against GRCh38 (no-alt analysis set + hs38d1 decoys). 2) Prioritisation: Exomiser 15.1.0 with data release 2602 (Ensembl 112 MANE transcripts, gnomAD v4.1, dbNSFP 4.5a, ClinVar 2026-02, HPO 2026-01), using the HPO terms supplied with the clinical phenotype (each verified against the current HPO release). Settings: PASS_ONLY; coding and splice-region variants only; max population frequency 2%; pathogenicity sources REVEL, MVP, AlphaMissense, SpliceAI; all inheritance modes considered, compound-heterozygous autosomal recessive weighted highest; hiPHIVE (human, mouse, zebrafish phenotypes + protein interaction network) and OMIM prioritisers. 3) Submission construction (bin/make_submission.py): recessive candidates submitted as compound-het pairs on one row, with each allele also listed singly as a hedge; pairs built only from recessive-mode variants at least 50 bp apart; exclusion of ClinVar-benign variants, hypervariable loci (HLA, KIR, MUC, olfactory receptors, Ig/TCR), lone heterozygous hits in recessive-only genes, and genes with Exomiser combined score < 0.1; documented gene-level curation (docs/curation.tsv). EPCR values are strictly descending and follow the Exomiser rank order; they are not calibrated probabilities. The output is validated against the scorer's format rules and every variant is checked to exist exactly as written in the normalised VCF. A parallel Ensembl VEP 116 annotation (AlphaMissense, ClinVar 2026-10) is produced for candidate review.
(required) If commercially-available Generative AI was used, record the provider, the plan or tier, and the relevant setting in your methods description. A line is enough, for example, "Anthropic API, Claude Sonnet, commercial terms, no training on customer content."	Anthropic, Claude via the claude.ai app (consumer Pro plan); model-training setting disabled ('Help improve Claude' off); no outputs rated. Used for pipeline design, code, troubleshooting and report drafting. No genome-scale data was shared with it.
Please state whether the submission file is the automated output of your computational approach (preferred), or if it has undergone downstream manual review and curation.	Both. The ranking, filtering, pairing and file generation are automated. A documented gene-level curation step (docs/curation.tsv) is then applied by the same script, so the final CSV is still script output and fully reproducible.

Please describe any downstream manual review in detail.	Each Exomiser candidate below the leading gene was reviewed for consequence, population frequency, ClinVar status, genotype and established disease association, and a decision with its reason recorded in docs/curation.tsv: • FANCD2 excluded: both variants classified benign/likely benign in ClinVar; pseudogene makes short-read calls unreliable. • HK1 excluded: intronic variant at 1.3% population frequency; heterozygous carrier state only. • ADAMTS1 excluded: two frameshifts with identical population frequency, suggesting one cis haplotype; no established Mendelian disease. • MTCH2 excluded: no established Mendelian disease; low phenotype score. • LZTR1 reclassified as a secondary finding (see Q on secondary findings). The review also led to the general rules applied automatically (no pairing < 50 bp; no lone recessive hits; ClinVar-benign exclusion), which removed further artefact 'pairs' (complex indels split into two records, repeat-length alleles, a segmentally duplicated gene). Rules and rationale are logged in docs/decisions.md.
Please state whether your approach only used publicly available data (preferred) or if proprietary data was also used. If proprietary data were used, where possible, please also include a submission based only on publicly available data.	Only publicly available data, in addition to the hackathon dataset. No proprietary data.
Please describe any public data used in detail (e.g., reference panels, population databases, literature).	• Reference genome: GRCh38 no-alt analysis set + hs38d1 decoys (NCBI). • Exomiser data 2602: Ensembl 112 (MANE) transcripts; gnomAD v4.1 exomes/genomes; dbSNP 156; UK10K; dbNSFP 4.5a (REVEL, MVP, AlphaMissense, SpliceAI); ClinVar 2026-02-08; OMIM 2026-02; Orphanet 2025-12; HPO and annotations 2026-01-08; IMPC/MGI mouse and ZFIN zebrafish phenotypes; STRING interaction network. • Ensembl VEP 116 cache and VEP_plugins release/116; AlphaMissense hg38 scores (Cheng et al. 2023); ClinVar GRCh38 VCF 2026-10-04 (NCBI). • Literature on MVA and BUB1B (e.g. Hanks et al., Nat Genet 2004).
Please describe any proprietary data used in detail.	None.
Is your approach able to output proposed pairs of compound heterozygous candidate variants, or only single candidate variants?	Yes. Compound-heterozygous candidates are output as pairs on one row (chrom_2/pos_2/ref_2/alt_2), with the individual alleles also listed as single rows below the pair to retain partial credit.
How did you handle secondary or incidental findings, if any? (See the finding_type column in your submission.)	Candidate genes unrelated to the phenotype but with a well-supported pathogenic variant are marked finding_type = secondary and placed below every primary row (EPCR 0.01), so they cannot act as false positives against the primary prediction. One secondary finding was reported: a heterozygous stop-gained variant in LZTR1, classified pathogenic/likely pathogenic in ClinVar; heterozygous LZTR1 loss-of-function is associated with schwannomatosis predisposition (incomplete penetrance). Unrelated to MVA.
Please provide an estimate of run time and cost for your approach (if possible).	On a 16-vCPU / 32 GB cloud server (Hetzner CPX62, about EUR 0.25/hour): normalisation ~30 s, Exomiser ~70 s, submission < 1 s; the optional VEP annotation ~1 h 45 min. End-to-end about 2 hours, under EUR 1 in compute. One-off setup (not included): ~125 GB of reference data and containers.

Provide a method abstract (up to 500 words). Please include strengths and limitations of your method as well as methodological detail.	We present a reproducible, phenotype-driven baseline for identifying the causal variant(s) in a single proband with Mosaic Variegated Aneuploidy (MVA), designed as a reference point for read-level analyses in later submissions. Methods. The provided GATK VCF (GRCh38) was restricted to primary chromosomes, filtered to PASS calls, split at multiallelic sites and left-normalised. Variants were prioritised with Exomiser 15.1.0 (data release 2602) using the supplied HPO terms, restricting to coding and splice-region variants with population frequency below 2%, combining REVEL, MVP, AlphaMissense and SpliceAI pathogenicity scores, and ranking genes by hiPHIVE phenotype similarity across human disease, mouse and zebrafish models and the protein interaction network. All inheritance modes were evaluated, with compound-heterozygous recessive inheritance weighted highest, since all known forms of MVA are recessive. A scripted step converted the ranking into the submission: recessive pairs on one row with single-allele hedges, exclusion of ClinVar-benign variants, hypervariable loci, variant 'pairs' closer than 50 bp (one event on one haplotype) and lone heterozygous recessive hits, followed by documented gene-level curation. The file is validated against the scorer's rules and every variant is checked against the source VCF. Results. Without any gene prior, Exomiser ranked BUB1B first, well ahead of the next gene. BUB1B encodes BUBR1, a core spindle assembly checkpoint kinase, and biallelic variants cause MVA type 1. The proband carries two heterozygous BUB1B variants: a stop-gained allele classified pathogenic/likely pathogenic in ClinVar, and an extremely rare missense allele not in ClinVar with a high AlphaMissense score (0.92). This truncating-plus-missense configuration matches the genotype pattern reported in MVA1, where complete loss of BUBR1 is not tolerated and the second allele is typically hypomorphic. The pair was submitted as the primary prediction; a heterozygous pathogenic LZTR1 stop-gain was reported as a secondary finding. Strengths. Fully reproducible (Nextflow, pinned containers, locked Python environment, versioned reference data); fast and cheap (about 2 hours and under EUR 1 of compute); transparent, with every exclusion rule and curation decision documented; compound-heterozygous output by design; general enough to apply to any singleton case. Limitations. Relies on the vendor's hard-filtered calls; phase of the two BUB1B variants is not established (singleton, no parental data, variants ~11 kb apart); only coding and splice-region variants were considered, so deep intronic, structural and mosaic variants are not assessed; the missense effect is predicted, not measured; EPCRs are rank-based rather than calibrated; curation involves expert judgement. Planned next steps address these: re-alignment from FASTQ and DeepVariant re-calling, read-backed phasing, mosaic-aware calling, structural variant and repeat-expansion calling, splice scoring of rare intronic variants, and read-depth / B-allele-frequency profiling to measure the patient's mosaic aneuploidy directly.